

# 3.2 Your First Program

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The C++ 20/23/26 Masterclass

## THE RUNNING EXAMPLE

Throughout this chapter we build **one small program**: a personal profile that greets you, asks a few questions, makes decisions, and - by the end - remembers your friends between runs.

- Nothing fancy on purpose
- One idea added per lecture, on top of the last
- The same excuse to meet every foundational building block

## BEFORE WE TYPE ANYTHING

[ SCREENSHOT ]

Every lecture folder has its own `CMakeLists.txt`, already scaffolded.  
Open the folder, don't create a new project by hand.

## THE SMALLEST REAL PROGRAM



```
#include <print>

int main() {

    std::println("Hello there!");
    std::println("Welcome to your very first C++ program.");

    return 0;
}
```

# WHAT'S ACTUALLY HAPPENING

| LINE                                | WHAT IT DOES                                   |
|-------------------------------------|------------------------------------------------|
| <code>#include &lt;print&gt;</code> | pulls in <code>std::println</code>             |
| <code>int main()</code>             | every C++ program starts here                  |
| <code>std::println(...)</code>      | prints text, adds the newline for you          |
| <code>return 0;</code>              | tells the OS the program finished successfully |

### 3.2 / RUN IT

## BUILD AND RUN

Same three commands every lecture in this course:



```
cmake -B build -G Ninja .  
cmake --build build  
./build/rooster
```

### 3.2 / RUN IT

## WHAT YOU SHOULD SEE

[ SCREENSHOT ]

## TWO KINDS OF COMMENTS



```
// A one-line comment - the compiler ignores everything after //
```

```
/*
```

```
    A block comment.
```

```
    Can span multiple lines - handy for a longer explanation.
```

```
*/
```

**Rule of thumb:** a comment earns its place by explaining **why**, not by narrating what the code obviously already says.



# Next: 3.3

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Statements and Functions